

Q1. Let α and β be the roots of $x^2 + \sqrt{3x} - 16 = 0$, and γ and δ be the roots of $x^2 + 3x - 1 = 0$. If $P_n = \alpha^n + \beta^n$ and $Q_n = \gamma^n + \delta^n$, then $\frac{P_{25} + \sqrt{3}P_{24}}{2P_{23}} + \frac{Q_{25} - Q_{23}}{Q_{24}}$ is equal to

(1) 3

(2) 4

(3) 5

(4) 7

Q2. For $t > -1$, let α_t and β_t be the roots of the equation $\left((t+2)^{\frac{1}{7}} - 1\right)x^2 + \left((t+2)^{\frac{1}{6}} - 1\right)x + \left((t+2)^{\frac{1}{21}} - 1\right) = 0$

If $\lim_{t \rightarrow -1^+} \alpha_t = a$ and $\lim_{t \rightarrow -1^+} \beta_t = b$, then $72(a+b)^2$ is equal to _____.

Q3. Let $P_n = \alpha^n + \beta^n, n \in \mathbf{N}$. If $P_{10} = 123, P_9 = 76, P_8 = 47$ and $P_1 = 1$, then the quadratic equation having roots $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ is :

(1) $x^2 - x + 1 = 0$

(2) $x^2 + x - 1 = 0$

(3) $x^2 - x - 1 = 0$

(4) $x^2 + x + 1 = 0$

Q4. If the set of all $a \in \mathbf{R} - \{1\}$, for which the roots of the equation $(1-a)x^2 + 2(a-3)x + 9 = 0$ are positive is $(-\infty, -\alpha] \cup [\beta, \gamma)$, then $2\alpha + \beta + \gamma$ is equal to _____.

Q5. Consider the equation $x^2 + 4x - n = 0$, where $n \in [20, 100]$ is a natural number. Then the number of all distinct values of n , for which the given equation has integral roots, is equal to

(1) 7

(2) 8

(3) 6

(4) 5

Q6. Let the set of all values of $p \in \mathbf{R}$, for which both the roots of the equation $x^2 - (p+2)x + (2p+9) = 0$ are negative real numbers, be the interval $(\alpha, \beta]$. Then $\beta - 2\alpha$ is equal to

(1) 0

(2) 9

(3) 5

(4) 20

Q7. The number of real roots of the equation $x|x-2| + 3|x-3| + 1 = 0$ is :

(1) 4

(2) 2

(3) 1

(4) 3

Q8. The sum of the squares of the roots of $|x+2|^2 + |x-2| - 2 = 0$ and the squares of the roots of $x^2 - 2|x-3| - 5 = 0$, is

(1) 26

(2) 36

(3) 30

(4) 24

Q1. Let $z \in \mathbb{C}$ be such that $\frac{z^2+3i}{z-2+i} = 2+3i$. Then the sum of all possible values of z^2 is

- (1) $19 - 2i$
- (2) $-19 - 2i$
- (3) $19 + 2i$
- (4) $-19 + 2i$

Q2. Let the product of $\omega_1 = (8+i)\sin\theta + (7+4i)\cos\theta$ and $\omega_2 = (1+8i)\sin\theta + (4+7i)\cos\theta$ be $\alpha + i\beta$, $i = \sqrt{-1}$. Let p and q be the maximum and the minimum values of $\alpha + \beta$ respectively.

- (1) 140
- (2) 130
- (3) 160
- (4) 150

Q3. Among the statements

(S1) : The set $\{z \in \mathbb{C} - \{-i\} : |z| = 1 \text{ and } \frac{z-i}{z+i} \text{ is purely real}\}$ contains exactly two elements, and (S2) : The set $\{z \in \mathbb{C} - \{-1\} : |z| = 1 \text{ and } \frac{z-1}{z+1} \text{ is purely imaginary}\}$ contains infinitely many elements.

- (1) both are incorrect
- (2) only (S1) is correct
- (3) only (S2) is correct
- (4) both are correct

Q4. Let $A =$

$$\left\{ \theta \in [0, 2\pi] : 1 + 10 \operatorname{Re} \left(\frac{2 \cos \theta + i \sin \theta}{\cos \theta - 3i \sin \theta} \right) = 0 \right\}.$$

Then $\sum_{\theta \in A} \theta^2$ is equal to

- (1) $\frac{21}{4}\pi^2$
- (2) $8\pi^2$
- (3) $\frac{27}{4}\pi^2$
- (4) $6\pi^2$

Q5. Let $A = \{z \in \mathbb{C} : |z - 2 - i| = 3\}$, $B = \{z \in \mathbb{C} : \operatorname{Re}(z - iz) = 2\}$ and $S = A \cap B$. Then $\sum_{z \in S} |z|^2$ is equal to

_____.

Q6. Let z be a complex number such that $|z| = 1$. If $\frac{2+k^2z}{k+z} = kz$, $k \in \mathbf{R}$, then the maximum distance of $k + ik^2$ from the circle $|z - (1 + 2i)| = 1$ is:

(1) $\sqrt{5} + 1$

(2) 2

(3) 3

(4) $\sqrt{3} + 1$

Q7. If $z_1, z_2, z_3 \in \mathbf{C}$ are the vertices of an equilateral triangle, whose centroid is z_0 , then $\sum_{k=1}^3 (z_k - z_0)^2$ is equal to

(1) 0

(2) 1

(3) i

(4) $-i$

Q8. If the locus of $z \in \mathbf{C}$, such that

$$\operatorname{Re}\left(\frac{z-1}{2z+i}\right) + \operatorname{Re}\left(\frac{\bar{z}-1}{2\bar{z}-i}\right) = 2$$

is a circle of radius r and center (a, b) then $\frac{15ab}{r^2}$ is equal to :

(1) 24

(2) 12

(3) 18

(4) 16

Q9. If α is a root of the equation $x^2 + x + 1 = 0$ and $\sum_{k=1}^n \left(\alpha^k + \frac{1}{\alpha^k}\right)^2 = 20$, then n is equal to

Q1. Let $a_1, a_2, a_3, \dots$ be in an A.P. such that $\sum_{k=1}^{12} a_{2k-1} = -\frac{72}{5}a_1, a_1 \neq 0$. If $\sum_{k=1}^n a_k = 0$, then n is:

(1) 11

(2) 10

(3) 18

(4) 17

Q2. Let a_n be the n^{th} term of an A. P.

If $S_n = a_1 + a_2 + a_3 + \dots + a_n = 700, a_6 = 7$ and $S_7 = 7$, then a_n is equal to :

(1) 56

(2) 65

(3) 64

(4) 70

Q3. Let $a_1, a_2, a_3, \dots$ be a G. P. of increasing positive numbers. If $a_3 a_5 = 729$ and $a_2 + a_4 = \frac{111}{4}$, then

$24(a_1 + a_2 + a_3)$ is equal to

(1) 131

(2) 130

(3) 129

(4) 128

Q4. The sum $1 + \frac{1+3}{2!} + \frac{1+3+5}{3!} + \frac{1+3+5+7}{4!} + \dots$ upto ∞ terms, is equal to

(1) $6e$

(2) $4e$

(3) $3e$

(4) $2e$

Q5. The number of terms of an A.P. is even; the sum of all the odd terms is 24, the sum of all the even terms is 30 and the last term exceeds the first by $\frac{21}{2}$. Then the number of terms which are integers in the A.P. is :

(1) 4

(2) 10

(3) 6

(4) 8

Q6. Consider two sets A and B , each containing three numbers in A.P. Let the sum and the product of the elements of A be 36 and p respectively and the sum and the product of the elements of B be 36 and q respectively. Let d and D be the common differences of AP's in A and B respectively such that $D = d + 3, d > 0$. If $\frac{p+q}{p-q} = \frac{19}{5}$, then $p - q$ is equal to

(1) 600

(2) 450

(3) 630

(4) 540

Q7. Let x_1, x_2, x_3, x_4 be in a geometric progression. 2, 7, 9, 5 are subtracted respectively from x_1, x_2, x_3, x_4 then the resulting numbers are in an arithmetic progression. Then the value of $\frac{1}{24}(x_1 x_2 x_3 x_4)$ is :

(1) 72

(2) 18

(3) 36

(4) 216

Q8. If the sum of the second, fourth and sixth terms of a G.P. of positive terms is 21 and the sum of its eighth, tenth and twelfth terms is 15309, then the sum of its first nine terms is :

(1) 760

(2) 755

(3) 750

(4) 757

Q9. If the sum of the first 10 terms of the series $\frac{4.1}{1+4.1^4} + \frac{4.2}{1+4.2^4} + \frac{4.3}{1+4.3^4} + \dots$ is $\frac{m}{n}$, where $\gcd(m, n) = 1$, then $m + n$ is equal to _____

Q10. The sum $1 + 3 + 11 + 25 + 45 + 71 + \dots$ upto 20 terms, is equal to

(1) 7240

(2) 7130

(3) 6982

(4) 8124

Q11. $1 + 3 + 5^2 + 7 + 9^2 + \dots$ upto 40 terms is equal to

(1) 43890

(2) 41880

(3) 33980

(4) 40870

Q12. If the sum of the first 20 terms of the series

$$\frac{4.1}{4+3.1^2+1^4} + \frac{4.2}{4+3.2^2+2^4} + \frac{4.3}{4+3.3^2+3^4} + \frac{4.4}{4+3.4^2+4^4} + \dots$$

is $\frac{m}{n}$, where m and n are coprime, then $m + n$ is equal to :-

(1) 423

(2) 420

(3) 421

(4) 422

Q13. Let $A = \{1, 6, 11, 16, \dots\}$ and $B = \{9, 16, 23, 30, \dots\}$ be the sets consisting of the first 2025 terms of two arithmetic progressions. Then $n(A \cup B)$ is

(1) 3814

(2) 4027

(3) 3761

(4) 4003

Q14. If $\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \dots \infty = \frac{\pi^4}{90}$,

$$\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots \infty = \alpha,$$

$$\frac{1}{2^4} + \frac{1}{4^4} + \frac{1}{6^4} + \dots \infty = \beta,$$

then $\frac{\alpha}{\beta}$ is equal to

(1) 23

(2) 18

Q1. The number of sequences of ten terms, whose terms are either 0 or 1 or 2, that contain exactly five 1's and exactly three 2's, is equal to

(1) 360

(2) 45

(3) 2520

(4) 1820

Q2. Line L_1 of slope 2 and line L_2 of slope $\frac{1}{2}$ intersect at the origin O. In the first quadrant, $P_1, P_2, \dots, P_{12}$ are 12 points on line L_1 and $Q_1, Q_2, \dots, Q_9$ are 9 points on line L_2 . Then the total number of triangles, that can be formed having vertices at three of the 22 points O, $P_1, P_2, \dots, P_{12}, Q_1, Q_2, \dots, Q_9$, is:

(1) 1080

(2) 1134

(3) 1026

(4) 1188

Q3. From a group of 7 batsmen and 6 bowlers, 10 players are to be chosen for a team, which should include at least 4 batsmen and at least 4 bowlers. One batsman and one bowler who are captain and vice-captain respectively of the team should be included. Then the total number of ways such a selection can be made, is

(1) 165

(2) 155

(3) 145

(4) 135

Q4. There are 12 points in a plane, no three of which are in the same straight line, except 5 points which are collinear. Then the total number of triangles that can be formed with the vertices at any three of these 12 points is

(1) 230

(2) 220

(3) 200

(4) 210

Q5. If the number of seven-digit numbers, such that the sum of their digits is even, is $m \cdot n \cdot 10^n$; $m, n \in \{1, 2, 3, \dots, 9\}$, then $m + n$ is equal to _____

Q6. Let p be the number of all triangles that can be formed by joining the vertices of a regular polygon P of n sides and q be the number of all quadrilaterals that can be formed by joining the vertices of P . If $p + q = 126$, then the eccentricity of the ellipse $\frac{x^2}{16} + \frac{y^2}{n} = 1$ is :

(1) $\frac{3}{4}$

(2) $\frac{1}{2}$

(3) $\frac{\sqrt{7}}{4}$

(4) $\frac{1}{\sqrt{2}}$

Q7. For $n \geq 2$, let S_n denote the set of all subsets of $\{1, 2, \dots, n\}$ with no two consecutive numbers. For example $\{1, 3, 5\} \in S_6$, but $\{1, 2, 4\} \notin S_6$. Then $n(S_5)$ is equal to _____

Q8. The largest $n \in \mathbb{N}$ such that 3^n divides $50!$ is:

(1) 21

(2) 22

(3) 20

(4) 23

Q9. Let m and n , ($m < n$) be two 2-digit numbers. Then the total numbers of pairs (m, n) , such that $\gcd(m, n) = 6$, is _____

Q10. The number of ways, in which the letters A, B, C, D, E can be placed in the 8 boxes of the figure below so that no row remains empty and at most one letter can be placed in a box, is :



(1) 5880

(2) 960

(3) 840

(4) 5760

Q1. The term independent of x in the expansion of $\left(\frac{(x+1)}{(x^{2/3}+1-x^{1/3})} - \frac{(x+1)}{(x-x^{1/2})}\right)^{10}$, $x > 1$ is:

(1) 210

(2) 150

(3) 240

(4) 120

Q2. The remainder when $((64)^{(64)})^{(64)}$ is divided by 7 is equal to

(1) 4

(2) 1

(3) 3

(4) 6

Q3. The sum of all rational terms in the expansion of $(2 + \sqrt{3})^8$ is

(1) 16923

(2) 3763

(3) 33845

(4) 18817

Q4. For an integer $n \geq 2$, if the arithmetic mean of all coefficients in the binomial expansion of $(x + y)^{2n-3}$ is 16, then the distance of the point $P(2n - 1, n^2 - 4n)$ from the line $x + y = 8$ is:

(1) $\sqrt{2}$

(2) $2\sqrt{2}$

(3) $5\sqrt{2}$

(4) $3\sqrt{2}$

Q5. In the expansion of $\left(\sqrt[3]{2} + \frac{1}{\sqrt[3]{3}}\right)^n$, $n \in \mathbb{N}$, if the ratio of 15th term from the beginning to the 15th term from the end is $\frac{1}{6}$, then the value of nC_3 is:

(1) 4060

(2) 1040

(3) 2300

(4) 4960

Q6. The product of the last two digits of $(1919)^{1919}$ is

Q7. The number of integral terms in the expansion of $\left(5^{\frac{1}{2}} + 7^{\frac{1}{8}}\right)^{1016}$ is

(1) 127

(2) 130

(3) 129

(4) 128

Q8. If $\sum_{r=0}^{10} \left(\frac{10^{r+1}-1}{10^r}\right) \cdot {}^{11}C_{r+1} = \frac{\alpha^{11}-11^{11}}{10^{10}}$, then α is equal to :

(1) 15

(2) 11

(3) 24

(4) 20

Q9. If $\sum_{r=1}^9 \left(\frac{r+3}{2^r}\right) \cdot {}^9C_r = \alpha \left(\frac{3}{2}\right)^9 - \beta$, $\alpha, \beta \in \mathbb{N}$, then $(\alpha + \beta)^2$ is equal to

(1) 27

(2) 9

(3) 81

(4) 18

Q10. Let $(1+x+x^2)^{10} = a_0 + a_1x + a_2x^2 + \dots + a_{20}x^{20}$. If $(a_1 + a_3 + a_5 + \dots + a_{19}) - 11a_2 = 121k$, then k is equal to .

Q11. If $1^2 \cdot ({}^{15}C_1) + 2^2 \cdot ({}^{15}C_2) + 3^2 \cdot ({}^{15}C_3) + \dots + 15^2 \cdot ({}^{15}C_{15}) = 2^m \cdot 3^n \cdot 5^k$, where $m, n, k \in \mathbb{N}$, then $m + n + k$ is equal to :

(1) 19

(2) 21

(3) 18

(4) 20

Q12. The sum of the series $2 \times 1 \times {}^{20}C_4 - 3 \times 2 \times {}^{20}C_5 + 4 \times 3 \times {}^{20}C_6 - 5 \times 4 \times {}^{20}C_7 + \dots + 18 \times 17 \times {}^{20}C_{20}$, is equal to

Q1. Let the mean and the standard deviation of the observation 2, 3, 3, 4, 5, 7, a, b be 4 and $\sqrt{2}$ respectively. Then the mean deviation about the mode of these observations is :

(1) 1

(2) $\frac{3}{4}$

(3) 2

(4) $\frac{1}{2}$

Q2. Let the Mean and Variance of five observations $x_1 = 1, x_2 = 3, x_3 = a, x_4 = 7$ and $x_5 = b, a > b$, be 5 and 10 respectively. Then the Variance of the observations $n + x_n, n = 1, 2, \dots, 5$ is

(1) 17

(2) 16.4

(3) 17.4

(4) 16

Q3. The mean and standard deviation of 100 observations are 40 and 5.1, respectively. By mistake one observation is taken as 50 instead of 40. If the correct mean and the correct standard deviation are μ and σ respectively, then $10(\mu + \sigma)$ is equal to

(1) 445

(2) 451

(3) 447

(4) 449

Q4. If the mean and the variance of 6, 4, a, 8, b, 12, 10, 13 are 9 and 9.25 respectively, then $a + b + ab$ is equal to :

(1) 105

(2) 103

(3) 100

(4) 106

Q1. Let $A = \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix}$. If for some $\theta \in (0, \pi)$, $A^2 = A^T$, then the sum of the diagonal elements of the matrix $(A + I)^3 + (A - I)^3 - 6A$ is equal to _____.

Q2. The number of singular matrices of order 2, whose elements are from the set $\{2, 3, 6, 9\}$ is _____.

Q3. Let $A = \begin{bmatrix} \alpha & -1 \\ 6 & \beta \end{bmatrix}$, $\alpha > 0$, such that $\det(A) = 0$ and $\alpha + \beta = 1$. If I denotes 2×2 identity matrix, then the matrix $(1 + A)^8$ is:

(1) $\begin{bmatrix} 4 & -1 \\ 6 & -1 \end{bmatrix}$

(2) $\begin{bmatrix} 257 & -64 \\ 514 & -127 \end{bmatrix}$

(3) $\begin{bmatrix} 1025 & -511 \\ 2024 & -1024 \end{bmatrix}$

(4) $\begin{bmatrix} 766 & -255 \\ 1530 & -509 \end{bmatrix}$

Q4. Let α be a solution of $x^2 + x + 1 = 0$, and for some a and b in

$\mathbb{R}$, $\begin{bmatrix} 1 & 16 & 13 \\ -1 & -1 & 2 \\ -2 & -14 & -8 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$. If $\frac{4}{\alpha^4} + \frac{m}{\alpha^a} + \frac{n}{\alpha^b} = 3$, then $m + n$ is equal to

(1) 3

(2) 11

(3) 7

(4) 8

Q5. Let A be a 3×3 real matrix such that $A^2(A - 2I) - 4(A - I) = O$, where I and O are the identity and null matrices, respectively. If $A^5 = \alpha A^2 + \beta A + \gamma I$, where α, β and γ are real constants, then $\alpha + \beta + \gamma$ is equal to:

(1) 12

(2) 20

(3) 76

(4) 4

Q6. Let the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ satisfy $A^n = A^{n-2} + A^2 - I$ for $n \geq 3$. Then the sum of all the elements of A^{50} is :-

(1) 53

(2) 52

(3) 39

(4) 44

Q7. Let $a \in \mathbf{R}$ and A be a matrix of order 3×3 such that $\det(A) = -4$ and $A + I = \begin{bmatrix} 1 & a & 1 \\ 2 & 1 & 0 \\ a & 1 & 2 \end{bmatrix}$, where I is the identity matrix of order 3×3 .

If $\det((a+1) \operatorname{adj}((a-1)A))$ is $2^m 3^n$, $m, n \in \{0, 1, 2, \dots, 20\}$, then $m + n$ is equal to :

(1) 14

(2) 17

(3) 15

(4) 16

Q8. Let A be a matrix of order 3×3 and $|A| = 5$. If $|2 \operatorname{adj}(3A \operatorname{adj}(2A))| = 2^\alpha \cdot 3^\beta \cdot 5^\gamma$, $\alpha, \beta, \gamma \in \mathbf{N}$ then $\alpha + \beta + \gamma$ is equal to

(1) 25

(2) 26

(3) 27

(4) 28

Q9. Let I be the identity matrix of order 3×3 and for the matrix $A = \begin{bmatrix} \lambda & 2 & 3 \\ 4 & 5 & 6 \\ 7 & -1 & 2 \end{bmatrix}$, $|A| = -1$. Let B be the inverse of the matrix $\operatorname{adj}(A \operatorname{adj}(A^2))$. Then $|(\lambda B + I)|$ is equal to _____

Q10. Let A be a 3×3 matrix such that $|\operatorname{adj}(\operatorname{adj}(\operatorname{adj} A))| = 81$. If

$$S = \left\{ n \in \mathbf{Z} : (|\operatorname{adj}(\operatorname{adj} A)|)^{\frac{(n-1)^2}{2}} = |A|^{(3n^2-5n-4)} \right\}$$

, then $\sum_{n \in S} |A^{(n^2+n)}|$ is equal to

(1) 866

(2) 750

(3) 820

(4) 732

Q11.

$$\text{Let } A = \begin{bmatrix} 2 & 2+p & 2+p+q \\ 4 & 6+2p & 8+3p+2q \\ 6 & 12+3p & 20+6p+3q \end{bmatrix}.$$

If $\det(\text{adj}(\text{adj}(3A))) = 2^m \cdot 3^n$, $m, n \in \mathbb{N}$, then $m + n$ is equal to

(1) 22

(2) 24

(3) 26

(4) 20

Q1. Let the system of equations

$$x + 5y - z = 1$$

$$4x + 3y - 3z = 7$$

$$24x + y + \lambda z = \mu$$

$\lambda, \mu \in \mathbb{R}$, have infinitely many solutions. Then the number of the solutions of this system, If x, y, z are integers and satisfy $7 \leq x + y + z \leq 77$, is

(1) 3

(2) 6

(3) 5

(4) 4

Q2.

If $y(x) = \begin{vmatrix} \sin x & \cos x & \sin x + \cos x + 1 \\ 27 & 28 & 27 \\ 1 & 1 & 1 \end{vmatrix}$, $x \in \mathbb{R}$, then $\frac{d^2y}{dx^2} + y$ is equal to

(1) -1

(2) 28

(3) 27

(4) 1

Q3. If the system of linear equations

$$3x + y + \beta z = 3$$

$$2x + \alpha y - z = -3$$

$$x + 2y + z = 4$$

has infinitely many solutions, then the value of $22\beta - 9\alpha$ is :

(1) 49

(2) 31

(3) 43

(4) 37

Q4. If the system of equation

$$2x + \lambda y + 3z = 5$$

$$3x + 2y - z = 7$$

$$4x + 5y + \mu z = 9$$

has infinitely many solutions, then $(\lambda^2 + \mu^2)$ is equal to :

(1) 22

(2) 18

(3) 26

(4) 30

Q5. Let the system of equations :

$$2x + 3y + 5z = 9$$

$$7x + 3y - 2z = 8$$

$$12x + 3y - (4 + \lambda)z = 16 - \mu$$

have infinitely many solutions. Then the radius of the circle centred at (λ, μ) and touching the line $4x = 3y$ is

(1) $\frac{17}{5}$

(2) $\frac{7}{5}$

(3) 7

(4) $\frac{21}{5}$

Q1. A card from a pack of 52 cards is lost. From the remaining 51 cards, n cards are drawn and are found to be spades. If the probability of the lost card to be a spade is $\frac{11}{50}$, the n is equal to

Q2. If the probability that the random variable X takes the value x is given by $P(X = x) = k(x + 1)3^{-x}$, $x = 0, 1, 2, 3, \dots$, where k is a constant, then $P(X \geq 3)$ is equal to

(1) $\frac{7}{27}$

(2) $\frac{4}{9}$

(3) $\frac{8}{27}$

(4) $\frac{1}{9}$

Q3. The probability, of forming a 12 persons committee from 4 engineers, 2 doctors and 10 professors containing at least 3 engineers and at least 1 doctor, is:

(1) $\frac{129}{182}$

(2) $\frac{103}{182}$

(3) $\frac{17}{26}$

(4) $\frac{19}{26}$

Q4. Given three indential bags each containing 10 balls, whose colours are as follows :

	Red	Blue	Green
Bag I	3	2	5
Bag II	4	3	3
Bag III	5	1	4

A person chooses a bag at random and takes out a ball. If the ball is Red, the probability that it is from bag I is p and if the balls is Green, the probability that it is from bag III is q , then the value of $\left(\frac{1}{p} + \frac{1}{q}\right)$ is :

(1) 6

(2) 9

(3) 7

(4) 8

Q5. A bag contains 19 unbiased coins and one coin with head on both sides. One coin drawn at random is tossed and head turns up. If the probability that the drawn coin was unbiased, is $\frac{m}{n}$, $\gcd(m, n) = 1$, then $n^2 - m^2$ is equal to :

(1) 80

(2) 60

(3) 72

(4) 64

Q6. A box contains 10 pens of which 3 are defective. A sample of 2 pens is drawn at random and let X denote the number of defective pens. Then the variance of X is

(1) $\frac{11}{15}$ (2) $\frac{28}{75}$ (3) $\frac{2}{15}$ (4) $\frac{3}{5}$

Q7. Let a random variable X take values 0, 1, 2, 3 with $P(X = 0) = P(X = 1) = p$, $P(X = 2) = P(X = 3)$ and $E(X^2) = 2E(X)$. Then the value of $8p - 1$ is :

(1) 0

(2) 2

(3) 1

(4) 3

Q8. Three distinct numbers are selected randomly from the set $\{1, 2, 3, \dots, 40\}$. If the probability, that the selected numbers are in an increasing G.P. is $\frac{m}{n}$, $\gcd(m, n) = 1$, then $m + n$ is equal to _____.

Q9. All five letter words are made using all the letters A, B, C, D, E and arranged as in an English dictionary with serial numbers. Let the word at serial number n be denoted by W_n . Let the probability $P(W_n)$ of choosing the word W_n satisfy $P(W_n) = 2P(W_{n-1})$, $n > 1$.

If $P(\text{CDBEA}) = \frac{2^\alpha}{2^\beta - 1}$, $\alpha, \beta \in \mathbb{N}$, then $\alpha + \beta$ is equal to : _____

Q10. If A and B are two events such that $P(A) = 0.7$, $P(B) = 0.4$ and $P(A \cap \bar{B}) = 0.5$, where $\bar{B}$ denotes the complement of B , then $P(B | (A \cup \bar{B}))$ is equal:-

(1) $\frac{1}{4}$ (2) $\frac{1}{2}$ (3) $\frac{1}{6}$ (4) $\frac{1}{3}$

Q1. Let $A = \{-2, -1, 0, 1, 2, 3\}$. Let R be a relation on A defined by xRy if and only if $y = \max\{x, 1\}$. Let l be the number of elements in R . Let m and n be the minimum number of elements required to be added in R to make it reflexive and symmetric relations, respectively. Then $l + m + n$ is equal to

(1) 12

(2) 11

(3) 13

(4) 14

Q2. Let $A = \{1, 2, 3, \dots, 10\}$ and R be a relation on A such that $R = \{(a, b) : a = 2b + 1\}$. Let $(a_1, a_2), (a_2, a_3), (a_3, a_4), \dots, (a_k, a_{k+1})$ be a sequence of k elements of R such that the second entry of an ordered pair is equal to the first entry of the next ordered pair. Then the largest integer k , for which such a sequence exists, is equal to :

(1) 6

(2) 7

(3) 5

(4) 8

Q3. Let A be the set of all functions $f : \mathbf{Z} \rightarrow \mathbf{Z}$ and R be a relation on A such that $R = \{(f, g) : f(0) = g(1) \text{ and } f(1) = g(0)\}$. Then R is:

(1) Symmetric and transitive but not reflexive

(2) Symmetric but neither reflexive nor transitive

(3) Reflexive but neither symmetric nor transitive

(4) Transitive but neither reflexive nor symmetric

Q4. Let $A = \{-3, -2, -1, 0, 1, 2, 3\}$. Let R be a relation on A defined by xRy if and only if $0 \leq x^2 + 2y \leq 4$.

Let l be the number of elements in R and m be the minimum number of elements required to be added in R to make it a reflexive relation. then $l + m$ is equal to

(1) 19

(2) 20

(3) 17

(4) 18

Q5. Let $A = \{-3, -2, -1, 0, 1, 2, 3\}$ and R be a relation on A defined by xRy if and only if $2x - y \in \{0, 1\}$. Let l be the number of elements in R . Let m and n be the minimum number of elements required to be added in R to make it reflexive and symmetric relations, respectively. Then $l + mn$ is equal to :-

(1) 18

(2) 17

(3) 15

(4) 16

Q6. The number of relations on the set $A = \{1, 2, 3\}$ containing at most 6 elements including $(1, 2)$, which are reflexive and transitive but not symmetric, is _____

Q7. Let $A = \{0, 1, 2, 3, 4, 5\}$. Let R be a relation on A defined by $(x, y) \in R$ if and only if $\max\{x, y\} \in \{3, 4\}$. Then among the statements (S_1) : The number of elements in R is 18, and (S_2) : The relation R is symmetric but neither reflexive nor transitive

(1) both are true

(2) both are false

(3) only (S_2) is true

(4) only (S_1) is true

Q1. Let f be a function such that $f(x) + 3f\left(\frac{24}{x}\right) = 4x, x \neq 0$. Then $f(3) + f(8)$ is equal to

- (1) 11
- (2) 10
- (3) 12
- (4) 13

Q2. Let $f(x) + 2f\left(\frac{1}{x}\right) = x^2 + 5$ and $2g(x) - 3g\left(\frac{1}{2}\right) = x, x > 0$. If $\alpha = \int_1^2 f(x)dx$, and $\beta = \int_1^2 g(x)dx$, then the value of $9\alpha + \beta$ is:

- (1) 1
- (2) 0
- (3) 10
- (4) 11

Q3. If the domain of the function

$$f(x) = \frac{1}{\sqrt{10+3x-x^2}} + \frac{1}{\sqrt{x+|x|}}$$

is (a, b) , then $(1+a)^2 + b^2$ is equal to :

- (1) 26
- (2) 29
- (3) 25
- (4) 30

Q4. If the domain of the function

$$f(x) = \log_e\left(\frac{2x-3}{5+4x}\right) + \sin^{-1}\left(\frac{4+3x}{2-x}\right)$$

is $[\alpha, \beta)$

then $\alpha^2 + 4\beta$ is equal to

- (1) 5
- (2) 4
- (3) 3
- (4) 7

Q5. If the domain of the function $f(x) = \log_7(1 - \log_4(x^2 - 9x + 18))$ is $(\alpha, \beta) \cup (\gamma, \delta)$, then $\alpha + \beta + \gamma + \delta$ is equal to

(1) 18

(2) 16

(3) 15

(4) 17

Q6. Let the domain of the function

$f(x) = \cos^{-1}\left(\frac{4x+5}{3x-7}\right)$ be $[\alpha, \beta]$ and the domain of $g(x) = \log_2(2 - 6\log_{27}(2x + 5))$ be (γ, δ) .

Then $|7(\alpha + \beta) + 4(\gamma + \delta)|$ is equal to _____

Q7. Let $f, g : (1, \infty) \rightarrow \mathbb{R}$ be defined as $f(x) = \frac{2x+3}{5x+2}$ and $g(x) = \frac{2-3x}{1-x}$. If the range of the function

$f \circ g : [2, 4] \rightarrow \mathbb{R}$ is $[\alpha, \beta]$, then $\frac{1}{\beta-\alpha}$ is equal to

(1) 68

(2) 29

(3) 2

(4) 56

Q8. Consider the sets $A = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x^2 + y^2 = 25\}$, $B = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x^2 + 9y^2 = 144\}$, $C = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : x^2 + y^2 \leq 4\}$, and $D = A \cap B$. The total number of one-one functions from the set D to the set C is:

(1) 15120

(2) 19320

(3) 17160

(4) 18290

Q9. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function satisfying $f(0) = 1$ and $f(2x) - f(x) = x$ for all $x \in \mathbb{R}$. If

$\lim_{n \rightarrow \infty} \left\{ f(x) - f\left(\frac{x}{2^n}\right) \right\} = G(x)$, then $\sum_{r=1}^{10} G(r^2)$ is equal to

(1) 540

(2) 385

(3) 420

(4) 215

Q10. Let the domains of the functions

$f(x) = \log_4 \log_3 \log_7(8 - \log_2(x^2 + 4x + 5))$ and $g(x) = \sin^{-1}\left(\frac{7x+10}{x-2}\right)$ be (α, β) and $[\gamma, \delta]$, respectively. Then

$\alpha^2 + \beta^2 + \gamma^2 + \delta^2$ is equal to :-

(1) 15

(2) 13

(3) 16

(4) 14

Q11. If the range of the function $f(x) = \frac{5-x}{x^2-3x+2}$, $x \neq 1, 2$, is $(-\infty, \alpha] \cup [\beta, \infty)$, then $\alpha^2 + \beta^2$ is equal to :

(1) 190

(2) 192

(3) 188

(4) 194

Q1. If the function $f(x) = \frac{\tan(\tan x) - \sin(\sin x)}{\tan x - \sin x}$ is continuous at $x = 0$, then $f(0)$ is equal to _____

Q2. If $\lim_{x \rightarrow 0} \frac{\cos(2x) + a \cos(4x) - b}{x^4}$ is finite, then $(a + b)$ is equal to :

(1) $\frac{1}{2}$

(2) 0

(3) $\frac{3}{4}$

(4) -1

Q3. $\lim_{x \rightarrow 0^+} \frac{\tan\left(5(x)^{\frac{1}{3}}\right) \log_e(1+3x^2)}{(\tan^{-1} 3\sqrt{x})^2 \left(e^{5(x)^{\frac{4}{3}}} - 1\right)}$ is equal to

(1) $\frac{1}{15}$

(2) 1

(3) $\frac{1}{3}$

(4) $\frac{5}{3}$

Q4. For $\alpha, \beta, \gamma, \in \mathbf{R}$, if $\lim_{x \rightarrow 0} \frac{x^2 \sin \alpha x + (\gamma - 1)e^{x^2}}{\sin 2x - \beta x} = 3$, then $\beta + \gamma - \alpha$ is equal to:

(1) 7

(2) 4

(3) 6

(4) -1

Q5. Given below are two statements :

Statement I : $\lim_{x \rightarrow 0} \left(\frac{\tan^{-1} x + \log_e \sqrt{\frac{1+x}{1-x}} - 2x}{x^5} \right) = \frac{2}{5}$

Statement II : $\lim_{x \rightarrow 1} \left(x^{\frac{2}{1-x}} \right) = \frac{1}{e^2}$

In the light of the above statements, choose the correct answer from the options given below :

(1) Statement I is false but Statement II is true

(2) Statement I is true but Statement II is false

(3) Both Statement I and Statement II are false

(4) Both Statement I and Statement II are true

Q6. If $\lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{\frac{1}{x^2}} = p$, then $96 \log_e p$ is equal to _____

Q7. If $\lim_{x \rightarrow 1^+} \frac{(x-1)(6+\lambda \cos(x-1))+\mu \sin(1-x)}{(x-1)^3} = -1$, where $\lambda, \mu \in \mathbb{R}$, then $\lambda + \mu$ is equal to

(1) 18

(2) 20

(3) 19

(4) 17

Q1. Let m and n be the number of points at which the function $f(x) = \max \{x, x^3, x^5, \dots, x^{21}\}$, $x \in \mathbb{R}$, is not differentiable and not continuous, respectively. Then $m + n$ is equal to _____.

Q2. Let $f(x) = \begin{cases} (1+ax)^{1/x} & , \quad x < 0 \\ 1+b & , \quad x = 0 \\ \frac{(x+4)^{1/2}-2}{(x+c)^{1/3}-2} & , \end{cases}$

be continuous at $x = 0$. Then e^{abc} is equal to

(1) 64

(2) 72

(3) 48

(4) 36

Q3. The number of points of discontinuity of the function $f(x) = \left\lceil \frac{x^2}{2} \right\rceil - [\sqrt{x}]$, $x \in [0, 4]$, where $[\cdot]$ denotes the greatest integer function is _____.

Q1. Let $f : \mathbf{R} \rightarrow \mathbf{R}$ be a twice differentiable function such that $(\sin x \cos y)(f(2x + 2y) - f(2x - 2y)) = (\cos x \sin y)(f(2x + 2y) + f(2x - 2y))$, for all $x, y \in \mathbf{R}$.

If $f'(0) = \frac{1}{2}$, then the value of $24f''\left(\frac{5\pi}{3}\right)$ is:

(1) 2

(2) -3

(3) 3

(4) -2

Q1. If the function $f(x) = 2x^3 - 9ax^2 + 12a^2x + 1$, where $a > 0$, attains its local maximum and local minimum values at p and q , respectively, such that $p^2 = q$, then $f(3)$ is equal to:

(1) 55

(2) 10

(3) 23

(4) 37

Q2. Let $x = -1$ and $x = 2$ be the critical points of the function $f(x) = x^3 + ax^2 + b \log_c |x| + 1$, $x \neq 0$. Let m and M respectively be the absolute minimum and the absolute maximum values of f in the interval $\left[-2, -\frac{1}{2}\right]$. Then

$|M + m|$ is equal to (Take $\log_c 2 = 0.7$):

(1) 21.1

(2) 19.8

(3) 22.1

(4) 20.9

Q3. The shortest distance between the curves $y^2 = 8x$ and $x^2 + y^2 + 12y + 35 = 0$ is :

(1) $2\sqrt{3} - 1$

(2) $\sqrt{2}$

(3) $3\sqrt{2} - 1$

(4) $2\sqrt{2} - 1$

Q4. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = \|x + 2\| - 2\|x\|$. If m is the number of points of local minima and n is the number of points of local maxima of f , then $m + n$ is

(1) 5

(2) 3

(3) 2

(4) 4

Q5. Let the function $f(x) = \frac{x}{3} + \frac{3}{x} + 3$, $x \neq 0$ be strictly increasing in $(-\infty, \alpha_1) \cup (\alpha_2, \infty)$ and strictly decreasing in $(\alpha_3, \alpha_4) \cup (\alpha_4, \alpha_5)$. Then $\sum_{i=1}^5 \alpha_i^2$ is equal to :-

(1) 48

(2) 28

(3) 40

(4) 36

Q6. Let $a > 0$. If the function $f(x) = 6x^3 - 45ax^2 + 108a^2x + 1$ attains its local maximum and minimum values at the points x_1 and x_2 respectively such that $x_1x_2 = 54$, then $a + x_1 + x_2$ is equal to :-

(1) 15

(2) 18

(3) 24

(4) 13

Q7. Let $f : \mathbf{R} \rightarrow \mathbf{R}$ be a polynomial function of degree four having extreme values at $x = 4$ and $x = 5$.

If $\lim_{x \rightarrow 0} \frac{f(x)}{x^2} = 5$, then $f(2)$ is equal to :

(1) 12

(2) 10

(3) 8

(4) 14

Q8. Let f be a differentiable function on $\mathbf{R}$ such that $f(2) = 1$, $f'(2) = 4$. Let $\lim_{x \rightarrow 0} (f(2+x))^{3/x} = e^\alpha$. Then the number of times the curve $y = 4x^3 - 4x^2 - 4(\alpha - 7)x - \alpha$ meets x -axis is :-

(1) 2

(2) 1

(3) 0

(4) 3

Q1. If $\int \frac{(\sqrt{1+x^2}+x)^{10}}{(\sqrt{1+x^2}-x)^9} dx =$

$\frac{1}{m} \left((\sqrt{1+x^2}+x)^n (n\sqrt{1+x^2}-x) \right) + C$ where C

is the constant of integration and $m, n \in \mathbb{N}$, then $m+n$ is equal to

Q2. Let $f(x) = \int x^3 \sqrt{3-x^2} dx$. If $5f(\sqrt{2}) = -4$, then $f(1)$ is equal to

(1) $-\frac{2\sqrt{2}}{5}$

(2) $-\frac{8\sqrt{2}}{5}$

(3) $-\frac{4\sqrt{2}}{5}$

(4) $-\frac{6\sqrt{2}}{5}$

Q3. $\int \left(\frac{1}{x} + \frac{1}{x^3} \right) \left(\sqrt[23]{3x^{-24} + x^{-26}} \right) dx$

If $= -\frac{\alpha}{3(\alpha+1)} (3x^\beta + x^\gamma)^{\frac{\alpha+1}{\alpha}} + C, x > 0,$ ($\alpha, \beta, \gamma \in \mathbb{Z}$), where C is the constant of

integration, then $\alpha + \beta + \gamma$ is equal to _____.

Q1. The integral $\int_0^{\pi} \frac{8x dx}{4 \cos^2 x + \sin^2 x}$ is equal to

(1) $2\pi^2$

(2) $4\pi^2$

(3) π^2

(4) $\frac{3\pi^2}{2}$

Q2. The integral $\int_0^{\pi} \frac{(x+3) \sin x}{1+3 \cos^2 x} dx$ is equal to :

(1) $\frac{\pi}{\sqrt{3}}(\pi + 1)$

(2) $\frac{\pi}{\sqrt{3}}(\pi + 2)$

(3) $\frac{\pi}{3\sqrt{3}}(\pi + 6)$

(4) $\frac{\pi}{2\sqrt{3}}(\pi + 4)$

Q3. Let $[.]$ denote the greatest integer function. If $\int_0^{e^3} \left[\frac{1}{e^x - 1} \right] dx = \alpha - \log_e 2$, then α^3 is equal to _____.

Q4. Let (a, b) be the point of intersection of the curve $x^2 = 2y$ and the straight line $y - 2x - 6 = 0$ in the second quadrant. Then the integral $I = \int_a^b \frac{9x^2}{1+5^x} dx$ is equal to :

(1) 24

(2) 27

(3) 18

(4) 21

Q5. $4 \int_0^1 \left(\frac{1}{\sqrt{3+x^2} + \sqrt{1+x^2}} \right) dx - 3 \log_e(\sqrt{3})$ is equal to :

(1) $2 + \sqrt{2} + \log_e(1 + \sqrt{2})$

(2) $2 - \sqrt{2} - \log_e(1 + \sqrt{2})$

(3) $2 + \sqrt{2} - \log_e(1 + \sqrt{2})$

(4) $2 - \sqrt{2} + \log_e(1 + \sqrt{2})$

Q6. Let the domain of the function

$f(x) = \log_2 \log_4 \log_6 (3 + 4x - x^2)$ be (a, b) . If $\int_0^{b-a} [x^2] dx = p - \sqrt{q} - \sqrt{r}$, $p, q, r \in \mathbb{N}$, $\gcd(p, q, r) = 1$,

where $[\cdot]$ is the greatest integer function, then $p + q + r$ is equal to

(1) 10

(2) 8

(3) 11

(4) 9

Q7. Let $f(x)$ be a positive function and $I_1 = \int_{-\frac{1}{2}}^1 2x f(2x(1-2x)) dx$ and $I_2 = \int_{-1}^2 f(x(1-x)) dx$. Then the value of

$\frac{I_2}{I_1}$ is equal to _____

(1) 9

(2) 6

(3) 12

(4) 4

Q8. The value of $\int_{-1}^1 \frac{(1+\sqrt{|x|-x})e^x + (\sqrt{|x|-x})e^{-x}}{e^x + e^{-x}} dx$ is equal to

(1) $3 - \frac{2\sqrt{2}}{3}$

(2) $2 + \frac{2\sqrt{2}}{3}$

(3) $1 - \frac{2\sqrt{2}}{3}$

(4) $1 + \frac{2\sqrt{2}}{3}$

Q9. The integral $\int_{-1}^{\frac{3}{2}} (|\pi^2 x \sin(\pi x)|) dx$ is equal to:

(1) $3 + 2\pi$

(2) $4 + \pi$

(3) $1 + 3\pi$

(4) $2 + 3\pi$

Q1. The area of the region $\{(x, y) : |x - y| \leq y \leq 4\sqrt{x}\}$ is

(1) 512

(2) $\frac{1024}{3}$

(3) $\frac{512}{3}$

(4) $\frac{2048}{3}$

Q2. If the area of the region

$$\{(x, y) : |4 - x^2| \leq y \leq x^2, y \leq 4, x \geq 0\}$$

is $\left(\frac{80\sqrt{2}}{\alpha} - \beta\right)$, $\alpha, \beta \in \mathbf{N}$, then $\alpha + \beta$ is equal to _____.

Q3. The area of the region bounded by the curve $y = \max\{|x|, x|x - 2|\}$, then x -axis and the lines $x = -2$ and $x = 4$ is equal to _____.

Q4. If the area of the region $\{(x, y) : |x - 5| \leq y \leq 4\sqrt{x}\}$ is A, then 3 A is equal to _____.

Q5. If the area of the region bounded by the curves $y = 4 - \frac{x^2}{4}$ and $y = \frac{x-4}{2}$ is equal to α , then 6α equals

(1) 250

(2) 210

(3) 240

(4) 220

Q6. If the area of the region $\{(x, y) : 1 + x^2 \leq y \leq \min\{x + 7, 11 - 3x\}\}$ is A, then 3 A is equal to

(1) 50

(2) 49

(3) 46

(4) 47

Q7. Let the area of the bounded region $\{(x, y) : 0 \leq 9x \leq y^2, y \geq 3x - 6\}$ be A. Then 6 A is equal to _____.

Q1. Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} + 3(\tan^2 x)y + 3y = \sec^2 x$
 $y(0) = \frac{1}{3} + e^3$. Then $y\left(\frac{\pi}{4}\right)$ is equal to

(1) $\frac{2}{3}$

(2) $\frac{4}{3}$

(3) $\frac{4}{3} + e^3$

(4) $\frac{2}{3} + e^3$

Q2. Let $f : \mathbf{R} \rightarrow \mathbf{R}$ be a thrice differentiable odd function satisfying $f'(x) \geq 0$, $f'(x) = f(x)$, $f(0) = 0$, $f'(0) = 3$.
Then $9f(\log_e 3)$ is equal to _____.

Q3. Let g be a differentiable function such that $\int_0^x g(t)dt = x - \int_0^x t g(t)dt$, $x \geq 0$ and let $y = y(x)$ satisfy the differential equation $\frac{dy}{dx} - y \tan x = 2(x+1) \sec x g(x)$, $x \in \left[0, \frac{\pi}{2}\right)$. If $y(0) = 0$, then $y\left(\frac{\pi}{3}\right)$ is equal to

(1) $\frac{2\pi}{3\sqrt{3}}$

(2) $\frac{4\pi}{3}$

(3) $\frac{2\pi}{3}$

(4) $\frac{4\pi}{3\sqrt{3}}$

Q4. Let $f : [0, \infty) \rightarrow \mathbf{R}$ be differentiable function such that $f(x) = 1 - 2x + \int_0^x e^{x-t} f(t)dt$ for all $x \in [0, \infty)$.
Then the area of the region bounded by $y = f(x)$ and the coordinate axes is

(1) $\sqrt{5}$

(2) $\frac{1}{2}$

(3) $\sqrt{2}$

(4) 2

Q5. If a curve $y = y(x)$ passes through the point $\left(1, \frac{\pi}{2}\right)$ and satisfies the differential equation

$(7x^4 \cot y - e^x \operatorname{cosec} y) \frac{dx}{dy} = x^5$, $x \geq 1$, then at $x = 2$, the value of $\cos y$ is:

(1) $\frac{2e^2 - e}{64}$

(2) $\frac{2e^2 + e}{64}$

(3) $\frac{2e^2 - e}{128}$

(4) $\frac{2e^2+e}{128}$

Q6. Let $f(x) = x - 1$ and $g(x) = e^x$ for $x \in \mathbb{R}$. If $\frac{dy}{dx} = \left(e^{-2\sqrt{x}} g(f(f(x))) - \frac{y}{\sqrt{x}} \right)$, $y(0) = 0$, then $y(1)$ is :-

(1) $\frac{1-e^2}{e^4}$

(2) $\frac{2e-1}{e^3}$

(3) $\frac{e-1}{e^4}$

(4) $\frac{1-e^3}{e^4}$

Q7. Let $f : [1, \infty) \rightarrow [2, \infty)$ be a differentiable function, If $10 \int_1^x f(t) dt = 5xf(x) - x^5 - 9$ for all $x \geq 1$, then the value of $f(3)$ is :

(1) 18

(2) 32

(3) 22

(4) 26

Q8. Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} + 2y \sec^2 x = 2 \sec^2 x + 3 \tan x \cdot \sec^2 x$ such that $y(0) = \frac{5}{4}$. Then $12 \left(y\left(\frac{\pi}{4}\right) - e^{-2} \right)$ is equal to _____.

Q9. Let $y = y(x)$ be the solution curve of the differential equation

$x(x^2 + e^x) dy + (e^x(x-2)y - x^3) dx = 0$, $x > 0$ passing through the point $(1, 0)$. Then $y(2)$ is equal to :

(1) $\frac{4}{4-e^2}$

(2) $\frac{2}{2+e^2}$

(3) $\frac{2}{2-e^2}$

(4) $\frac{4}{4+e^2}$

Q10. Let $y = y(x)$ be the solution of the differential equation $(x^2 + 1) y' - 2xy = (x^4 + 2x^2 + 1) \cos x$, $y(0) = 1$. Then $\int_{-3}^3 y(x) dx$ is :

(1) 24

(2) 36

(3) 30

- Q1.** Let $A(4, -2)$, $B(1, 1)$ and $C(9, -3)$ be the vertices of a triangle ABC . Then the maximum area of the parallelogram $AFDE$, formed with vertices D , E and F on the sides BC , CA and AB of the triangle ABC respectively, is _____.
- Q2.** Let the area of the triangle formed by a straight Line $L: x + by + c = 0$ with co-ordinate axes be 48 square units. If the perpendicular drawn from the origin to the line L makes an angle of 45° with the positive x -axis, then the value of $b^2 + c^2$ is:
- (1) 90
(2) 93
(3) 97
(4) 83
- Q3.** Let the equation $x(x + 2)(12 - k) = 2$ have equal roots. Then the distance of the point $\left(k, \frac{k}{2}\right)$ from the line $3x + 4y + 5 = 0$ is
- (1) 15
(2) $5\sqrt{3}$
(3) $15\sqrt{5}$
(4) 12
- Q4.** Let the three sides of a triangle are on the lines $4x - 7y + 10 = 0$, $x + y = 5$ and $7x + 4y = 15$. Then the distance of its orthocentre from the orthocentre of the triangle formed by the lines $x = 0$, $y = 0$ and $x + y = 1$ is
- (1) 5
(2) $\sqrt{5}$
(3) $\sqrt{20}$
(4) 20
- Q5.** Let ABC be the triangle such that the equations of lines AB and AC be $3y - x = 2$ and $x + y = 2$, respectively, and the points B and C lie on x -axis. If P is the orthocentre of the triangle ABC , then the area of the triangle PBC is equal to
- (1) 4
(2) 10
(3) 8

(4) 6

Q6. If the orthocentre of the triangle formed by the lines $y = x + 1$, $y = 4x - 8$ and $y = mx + c$ is at $(3, -1)$, then $m - c$ is :

(1) 0

(2) -2

(3) 4

(4) 2

Q7. A line passing through the point $P(a, \theta)$ makes an acute angle α with the positive x -axis. Let this line be rotated about the point P through an angle $\frac{\alpha}{2}$ in the clock-wise direction. If in the new position, the slope of the line is $2 - \sqrt{3}$ and its distance from the origin is $\frac{1}{\sqrt{2}}$, then the value of $3a^2 \tan^2 \alpha - 2\sqrt{3}$ is

(1) 4

(2) 6

(3) 5

(4) 8

Q8. A line passes through the origin and makes equal angles with the positive coordinate axes. It intersects the lines $L_1 : 2x + y + 6 = 0$ and $L_2 : 4x + 2y - p = 0$, $p > 0$, at the points A and B , respectively. If $AB = \frac{9}{\sqrt{2}}$ and the foot of the perpendicular from the point A on the line L_2 is M , then $\frac{AM}{BM}$ is equal to

(1) 5

(2) 4

(3) 2

(4) 3

Q9. Consider the lines $x(3\lambda + 1) + y(7\lambda + 2) = 17\lambda + 5$, λ being a parameter, all passing through a point P . One of these lines (say L) is farthest from the origin. If the distance of L from the point $(3, 6)$ is d , then the value of d^2 is

(1) 20

(2) 30

(3) 10

(4) 15

Q10. Let a be the length of a side of a square OABC with O being the origin. Its side OA makes an acute angle α with the positive x -axis and the equations of its diagonals are $(\sqrt{3} + 1)x + (\sqrt{3} - 1)y = 0$ and $(\sqrt{3} - 1)x - (\sqrt{3} + 1)y + 8\sqrt{3} = 0$. Then a^2 is equal to

(1) 48

(2) 32

(3) 16

(4) 24

Q1. The absolute difference between the squares of the radii of the two circles passing through the point $(-9, 4)$ and touching the lines $x + y = 3$ and $x - y = 3$, is equal to _____.

Q2. If the four distinct points $(4, 6)$, $(-1, 5)$, $(0, 0)$ and $(k, 3k)$ lie on a circle of radius r , then $10k + r^2$ is equal to

(1) 32

(2) 33

(3) 34

(4) 35

Q3. Let C_1 be the circle in the third quadrant of radius 3, that touches both coordinate axes. Let C_2 be the circle with centre $(1, 3)$ that touches C_1 externally at the point (α, β) . If $(\beta - \alpha)^2 = \frac{m}{n}$, $\gcd(m, n) = 1$, then $m + n$ is equal to :

(1) 9

(2) 13

(3) 22

(4) 31

Q1. The axis of a parabola is the line $y = x$ and its vertex and focus are in the first quadrant at distances $\sqrt{2}$ and $2\sqrt{2}$ units from the origin, respectively. If the point $(1, k)$ lies on the parabola, then a possible value of k is :-

(1) 4

(2) 9

(3) 3

(4) 8

Q2. Let P be the parabola, whose focus is $(-2, 1)$ and directrix is $2x + y + 2 = 0$. Then the sum of the ordinates of the points on P , whose abscissa is -2 , is

(1) $\frac{3}{2}$

(2) $\frac{5}{2}$

(3) $\frac{1}{4}$

(4) $\frac{3}{4}$

Q3. Let the focal chord PQ of the parabola $y^2 = 4x$ make an angle of 60° with the positive x -axis, where P lies in the first quadrant. If the circle, whose one diameter is PS , S being the focus of the parabola, touches the y -axis at the point $(0, \alpha)$, then $5\alpha^2$ is equal to :

(1) 15

(2) 25

(3) 30

(4) 20

Q4. Let the point P of the focal chord PQ of the parabola $y^2 = 16x$ be $(1, -4)$. If the focus of the parabola divides the chord PQ in the ratio $m : n$, $\gcd(m, n) = 1$, then $m^2 + n^2$ is equal to :

(1) 17

(2) 10

(3) 37

(4) 26

Q5. The radius of the smallest circle which touches the parabolas $y = x^2 + 2$ and $x = y^2 + 2$ is

(1) $\frac{7\sqrt{2}}{2}$

(2) $\frac{7\sqrt{2}}{16}$

(3) $\frac{7\sqrt{2}}{4}$

(4) $\frac{7\sqrt{2}}{8}$

Q6. A line passing through the point $A(-2, 0)$, touches the parabola $P : y^2 = x - 2$ at the point B in the first quadrant. The area, of the region bounded by the line AB , parabola P and the x -axis, is :-

(1) $\frac{7}{3}$

(2) 2

(3) $\frac{8}{3}$

(4) 3

Q7. Let r be the radius of the circle, which touches x -axis at point $(a, 0)$, $a < 0$ and the parabola $y^2 = 9x$ at the point $(4, 6)$. Then r is equal to _____

Q1. If S and S' are the foci of the ellipse $\frac{x^2}{18} + \frac{y^2}{9} = 1$ and P be a point on the ellipse, then $\min (SP, S'P) + \max (SP, S'P)$ is equal to :

(1) $3(1 + \sqrt{2})$

(2) $3(6 + \sqrt{2})$

(3) 9

(4) 27

Q2. If the length of the minor axis of an ellipse is equal to one fourth of the distance between the foci, then the eccentricity of the ellipse is :

(1) $\frac{4}{\sqrt{17}}$

(2) $\frac{\sqrt{3}}{16}$

(3) $\frac{3}{\sqrt{19}}$

(4) $\frac{\sqrt{5}}{7}$

Q3. A line passing through the point $P(\sqrt{5}, \sqrt{5})$ intersects the ellipse $\frac{x^2}{36} + \frac{y^2}{25} = 1$ at A and B such that $(PA) \cdot (PB)$ is maximum. Then $5(PA^2 + PB^2)$ is equal to :

(1) 218

(2) 377

(3) 290

(4) 338

Q4. The length of the latus-rectum of the ellipse, whose foci are $(2, 5)$ and $(2, -3)$ and eccentricity is $\frac{4}{5}$, is

(1) $\frac{6}{5}$

(2) $\frac{50}{3}$

(3) $\frac{10}{3}$

(4) $\frac{18}{5}$

Q5. The centre of a circle C is at the centre of the ellipse $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b$. Let C pass through the foci F_1 and F_2 of E such that the circle C and the ellipse E intersect at four points. Let P be one of these four points. If the

area of the triangle PF_1F_2 is 30 and the length of the major axis of E is 17, then the distance between the foci of E is :

(1) 26

(2) 13

(3) 12

(4) $\frac{13}{2}$

Q6. Let

$A = \{(\alpha, \beta) \in \mathbf{R} \times \mathbf{R} : |\alpha + 1| \leq 4 \text{ and } |\beta - 5| \leq 6\}$
and

$B = \{(\alpha, \beta) \in \mathbf{R} \times \mathbf{R} : 16(\alpha - 2)^2 + 9(\beta - 6)^2 \leq 144\}$

(1) $B \subset A$

(2) $A \cup B = \{(x, y) : -4 \leq x \leq 4, -1 \leq y \leq 11\}$

(3) neither $A \subset B$ nor $B \subset A$

(4) $A \subset B$

Q7. Let the length of a latus rectum of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ be 10. If its eccentricity is the minimum value of the function $f(t) = t^2 + t + \frac{11}{12}$, $t \in \mathbf{R}$, then $a^2 + b^2$ is equal to :

(1) 125

(2) 126

(3) 120

(4) 115

Q8. Let the ellipse $3x^2 + py^2 = 4$ pass through the centre C of the circle $x^2 + y^2 - 2x - 4y - 11 = 0$ of radius r . Let f_1, f_2 be the focal distances of the point C on the ellipse. Then $6f_1f_2 - r$ is equal to

(1) 74

(2) 68

(3) 70

(4) 78

Q9. Let for two distinct values of p the lines $y = x + p$ touch the ellipse E : $\frac{x^2}{4^2} + \frac{y^2}{3^2} = 1$ at the points A and B. Let the line $y = x$ intersect E at the points C and D. Then the area of the quadrilateral ABCD is equal to

(1) 36

(2) 24

(3) 48

(4) 20

Q10. Let C be the circle of minimum area enclosing the ellipse $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with eccentricity $\frac{1}{2}$ and foci $(\pm 2, 0)$.

Let PQR be a variable triangle, whose vertex P is on the circle C and the side QR of length 29 is parallel to the major axis of E and contains the point of intersection of E with the negative y -axis. Then the maximum area of the triangle PQR is :

(1) $6(3 + \sqrt{2})$

(2) $8(3 + \sqrt{2})$

(3) $62 + \sqrt{3}$

(4) $82 + \sqrt{3}$

Q11. Let C be the circle $x^2 + (y - 1)^2 = 2$, E_1 and E_2 be two ellipses whose centres lie at the origin and major axes lie on x -axis and y -axis respectively. Let the straight line $x + y = 3$ touch the curves C , E_1 and E_2 at

$P(x_1, y_1)$, $Q(x_2, y_2)$ and $R(x_3, y_3)$ respectively. Given that P is the mid-point of the line segment QR and

$PQ = \frac{2\sqrt{2}}{3}$, the value of $9(x_1y_1 + x_2y_2 + x_3y_3)$ is equal to _____.

Q1. Let one focus of the hyperbola $H : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be at $(\sqrt{10}, 0)$ and the corresponding directrix be $x = \frac{9}{\sqrt{10}}$. If e and l respectively are the eccentricity and the length of the latus rectum of H , then $9(e^2 + l)$ is equal to:

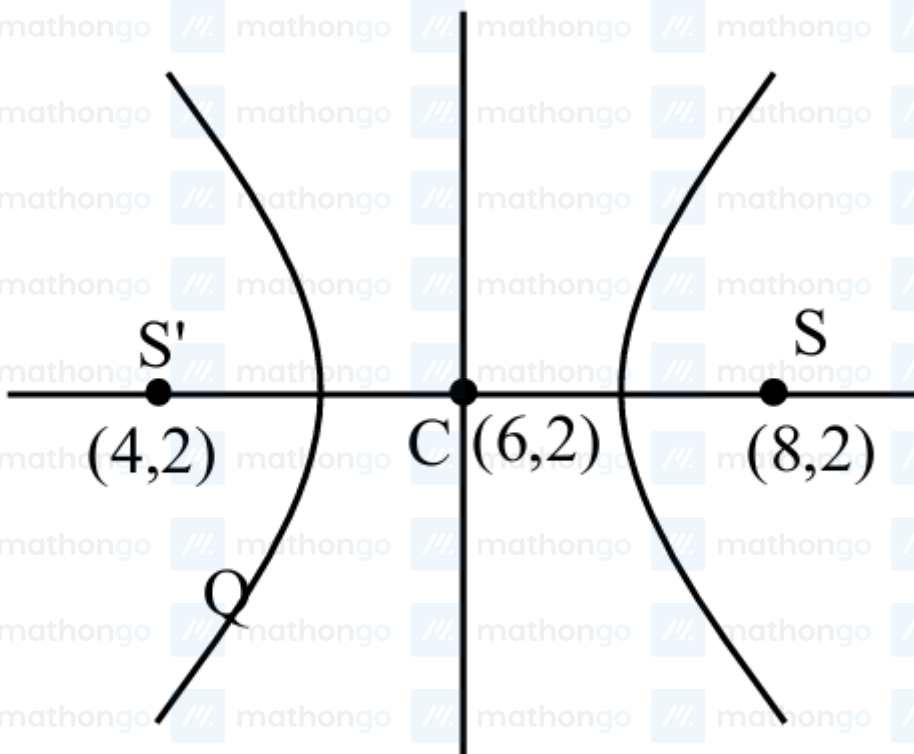
(1) 14

(2) 15

(3) 16

(4) 12

Q2. If the equation of the hyperbola with foci $(4, 2)$ and $(8, 2)$ is $3x^2 - y^2 - \alpha x + \beta y + \gamma = 0$, then $\alpha + \beta + \gamma$ is equal to _____.



Q3. Let e_1 and e_2 be the eccentricities of the ellipse $\frac{x^2}{b^2} + \frac{y^2}{25} = 1$ and the hyperbola $\frac{x^2}{16} - \frac{y^2}{b^2} = 1$, respectively. If $b < 5$ and $e_1 e_2 = 1$, then the eccentricity of the ellipse having its axes along the coordinate axes and passing through all four foci (two of the ellipse and two of the hyperbola) is :

(1) $\frac{4}{5}$

(2) $\frac{3}{5}$

(3) $\frac{\sqrt{7}}{4}$

(4) $\frac{\sqrt{3}}{2}$

Q4. Let the product of the focal distances of the point $P(4, 2\sqrt{3})$ on the hyperbola $H : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be 32 .

Let the length of the conjugate axis of H be p and the length of its latus rectum be q . Then $p^2 + q^2$ is equal to

Q5. Let the sum of the focal distances of the point $P(4, 3)$ on the hyperbola $H : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be $8\sqrt{\frac{5}{3}}$. If for H , the length of the latus rectum is l and the product of the focal distances of the point P is m , then $9l^2 + 6m$ is equal to :-

(1) 184

(2) 186

(3) 185

(4) 187

Q6. Consider the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ having one of its focus at $P(-3, 0)$. If the latus ractum through its other focus subtends a right angle at P and $a^2b^2 = \alpha\sqrt{2} - \beta$, $\alpha, \beta \in \mathbb{N}$.

Q7. Let the lengths of the transverse and conjugate axes of a hyperbola in standard form be $2a$ and $2b$, respectively, and one focus and the corresponding directrix of this hyperbola be $(-5, 0)$ and $5x + 9 = 0$, respectively. If the product of the focal distances of a point $(\alpha, 2\sqrt{5})$ on the hyperbola is p , then $4p$ is equal to

Q1. If $\theta \in \left[-\frac{7\pi}{6}, \frac{4\pi}{3}\right]$, then the number of solutions of $\sqrt{3} \operatorname{cosec}^2 \theta - 2(\sqrt{3} - 1) \operatorname{cosec} \theta - 4 = 0$, is equal to

(1) 6

(2) 8

(3) 10

(4) 7

Q2. If $\theta \in [-2\pi, 2\pi]$, then the number of solutions of $2\sqrt{2} \cos^2 \theta + (2 - \sqrt{6}) \cos \theta - \sqrt{3} = 0$, is equal to:

(1) 12

(2) 6

(3) 8

(4) 10

Q3. The number of solutions of the equation $2x + 3 \tan x = \pi, x \in [-2\pi, 2\pi] - \left\{\pm \frac{\pi}{2}, \pm \frac{3\pi}{2}\right\}$ is

(1) 6

(2) 5

(3) 4

(4) 3

Q4. The number of solutions of equation $(4 - \sqrt{3}) \sin x - 2\sqrt{3} \cos^2 x = -\frac{4}{1+\sqrt{3}}, x \in \left[-2\pi, \frac{5\pi}{2}\right]$ is

(1) 4

(2) 3

(3) 6

(4) 5

Q5. If $10 \sin^4 \theta + 15 \cos^4 \theta = 6$, then the value of $\frac{27 \operatorname{cosec}^6 \theta + 8 \sec^6 \theta}{16 \sec^8 \theta}$ is:

(1) $\frac{2}{5}$

(2) $\frac{3}{4}$

(3) $\frac{3}{5}$

(4) $\frac{1}{5}$

Q6. If for $\theta \in \left[-\frac{\pi}{3}, 0\right]$, the points $(x, y) = \left(3 \tan\left(\theta + \frac{\pi}{3}\right), 2 \tan\left(\theta + \frac{\pi}{6}\right)\right)$ lie on $xy + \alpha x + \beta y + \gamma = 0$, then $\alpha^2 + \beta^2 + \gamma^2$ is equal to :

(1) 80

(2) 72

(3) 96

(4) 75

Q7. The number of solutions of the equation

$$\cos 2\theta \cos \frac{\theta}{2} + \cos \frac{5\theta}{2} = 2 \cos^3 \frac{5\theta}{2}$$

in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ is :

(1) 7

(2) 5

(3) 6

(4) 7

Q1. The value of $\cot^{-1}\left(\frac{\sqrt{1+\tan^2(2)}-1}{\tan(2)}\right) - \cot^{-1}$

$\left(\frac{\sqrt{1+\tan^2\left(\frac{1}{2}\right)}+1}{\tan\left(\frac{1}{2}\right)}\right)$ is equal to

(1) $\pi - \frac{5}{4}$

(2) $\pi - \frac{3}{2}$

(3) $\pi + \frac{3}{2}$

(4) $\pi + \frac{5}{2}$

Q2. If $y = \cos\left(\frac{\pi}{3} + \cos^{-1}\frac{x}{2}\right)$, then $(x - y)^2 + 3y^2$ is equal to _____.

Q3. Considering the principal values of the inverse trigonometric functions,

$\sin^{-1}\left(\frac{\sqrt{3}}{2}x + \frac{1}{2}\sqrt{1-x^2}\right), -\frac{1}{2} < x < \frac{1}{\sqrt{2}}$, is equal to

(1) $\frac{\pi}{4} + \sin^{-1}x$

(2) $\frac{\pi}{6} + \sin^{-1}x$

(3) $\frac{-5\pi}{6} - \sin^{-1}x$

(4) $\frac{5\pi}{6} - \sin^{-1}x$

Q4. The sum of the infinite series

$\cot^{-1}\left(\frac{7}{4}\right) + \cot^{-1}\left(\frac{19}{4}\right) + \cot^{-1}\left(\frac{39}{4}\right) + \cot^{-1}\left(\frac{67}{4}\right) + \dots$ is :-

(1) $\frac{\pi}{2} + \tan^{-1}\left(\frac{1}{2}\right)$

(2) $\frac{\pi}{2} - \cot^{-1}\left(\frac{1}{2}\right)$

(3) $\frac{\pi}{2} + \cot^{-1}\left(\frac{1}{2}\right)$

(4) $\frac{\pi}{2} - \tan^{-1}\left(\frac{1}{2}\right)$

Q1. Let $ABCD$ be a tetrahedron such that the edges AB , AC and AD are mutually perpendicular. Let the areas of the triangles ABC , ACD and ADB be 5, 6 and 7 square units respectively. Then the area (in square units) of the $\triangle BCD$ is equal to :

(1) $\sqrt{340}$

(2) 12

(3) $\sqrt{110}$

(4) $7\sqrt{3}$

Q2. Let the angle θ , $0 < \theta < \frac{\pi}{2}$ between two unit vectors $\hat{a}$ and $\hat{b}$ be $\sin^{-1}\left(\frac{\sqrt{65}}{9}\right)$. If the vector

$\vec{c} = 3\hat{a} + 6\hat{b} + 9(\hat{a} \times \hat{b})$, then the value of $9(\vec{c} \cdot \hat{a}) - 3(\vec{c} \cdot \hat{b})$ is

(1) 31

(2) 27

(3) 29

(4) 24

Q3. Let the three sides of a triangle ABC be given by the vectors $2\hat{i} - \hat{j} + \hat{k}$, $\hat{i} - 3\hat{j} - 5\hat{k}$ and $3\hat{i} - 4\hat{j} - 4\hat{k}$. Let G be the centroid of the triangle ABC . Then $6\left(|\vec{AG}|^2 + |\vec{BG}|^2 + |\vec{CG}|^2\right)$ is equal to _____

Q4. Let $\vec{a}$ and $\vec{b}$ be the vectors of the same magnitude such that $\frac{|\vec{a}+\vec{b}|+|\vec{a}-\vec{b}|}{|\vec{a}+\vec{b}|-|\vec{a}-\vec{b}|} = \sqrt{2} + 1$. Then $\frac{|\vec{a}+\vec{b}|^2}{|\vec{a}|^2}$ is :

(1) $2 + 4\sqrt{2}$

(2) $1 + \sqrt{2}$

(3) $2 + \sqrt{2}$

(4) $4 + 2\sqrt{2}$

Q5. If $\vec{a}$ is nonzero vector such that its projections on the vectors $2\hat{i} - \hat{j} + 2\hat{k}$, $\hat{i} + 2\hat{j} - 2\hat{k}$ and $\hat{k}$ are equal, then a unit vector along $\vec{a}$ is:

(1) $\frac{1}{\sqrt{155}}(-7\hat{i} + 9\hat{j} + 5\hat{k})$

(2) $\frac{1}{\sqrt{155}}(-7\hat{i} + 9\hat{j} - 5\hat{k})$

(3) $\frac{1}{\sqrt{155}}(7\hat{i} + 9\hat{j} + 5\hat{k})$

(4) $\frac{1}{\sqrt{155}}(7\hat{i} + 9\hat{j} - 5\hat{k})$

Q6. Let $\vec{a} = 2\hat{i} - 3\hat{j} + \hat{k}$, $\vec{b} = 3\hat{i} + 2\hat{j} + 5\hat{k}$ and a vector $\vec{c}$ be such that $(\vec{a} - \vec{c}) \times \vec{b} = -18\hat{i} - 3\hat{j} + 12\hat{k}$ and $\vec{a} \cdot \vec{c} = 3$. If $\vec{b} \times \vec{c} = \vec{d}$, then $|\vec{a} \cdot \vec{d}|$ is equal to :

(1) 18

(2) 12

(3) 9

(4) 15

Q7. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = 3\hat{i} + 2\hat{j} - \hat{k}$, $\vec{c} = \lambda\hat{j} + \mu\hat{k}$ and $\hat{d}$ be a unit vector such that $\vec{a} \times \hat{d} = \vec{b} \times \hat{d}$ and $\vec{c} \cdot \hat{d} = 1$. If $\vec{c}$ is perpendicular to $\vec{a}$, then $|3\lambda\hat{d} + \mu\vec{c}|^2$ is equal to _____.

Q8. Let $\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$, $\vec{b} = 3\hat{i} - 3\hat{j} + 3\hat{k}$, $\vec{c} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{d}$ be a vector such that $\vec{b} \times \vec{d} = \vec{c} \times \vec{d}$ and $\vec{a} \cdot \vec{d} = 4$. Then $|\vec{a} \times \vec{d}|^2$ is equal to _____.

Q9. Consider two vectors $\vec{u} = 3\hat{i} - \hat{j}$ and $\vec{v} = 2\hat{i} + \hat{j} - \lambda\hat{k}$, $\lambda > 0$. The angle between them is given by $\cos^{-1}\left(\frac{\sqrt{5}}{2\sqrt{7}}\right)$. Let $\vec{v} = \vec{v}_1 + \vec{v}_2$, where $\vec{v}_1$ is parallel to $\vec{u}$ and $\vec{v}_2$ is perpendicular to $\vec{u}$. Then the value $|\vec{v}_1|^2 + |\vec{v}_2|^2$ is equal to

(1) $\frac{23}{2}$

(2) 14

(3) $\frac{25}{2}$

(4) 10

Q10. Let $\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} + \hat{j} - \hat{k}$. Let $\hat{c}$ be a unit vector in the plane of the vectors $\vec{a}$ and $\vec{b}$ and be perpendicular to $\vec{a}$. Then such a vector $\hat{c}$ is :

(1) $\frac{1}{\sqrt{5}}(\hat{j} - 2\hat{k})$

(2) $\frac{1}{\sqrt{3}}(-\hat{i} + \hat{j} - \hat{k})$

(3) $\frac{1}{\sqrt{3}}(\hat{i} - \hat{j} + \hat{k})$

(4) $\frac{1}{\sqrt{2}}(-\hat{i} + \hat{k})$

Q1. Each of the angles β and γ that a given line makes with the positive y - and z -axes, respectively, is half of the angle that this line makes with the positive x -axes. Then the sum of all possible values of the angle β is

(1) $\frac{3\pi}{4}$

(2) π

(3) $\frac{\pi}{2}$

(4) $\frac{3\pi}{2}$

Q2. The line L_1 is parallel to the vector $\vec{a} = -3\hat{i} + 2\hat{j} + 4\hat{k}$ and passes through the point $(7, 6, 2)$ and the line L_2 is parallel to the vector $\vec{b} = 2\hat{i} + \hat{j} + 3\hat{k}$ and passes through the point $(5, 3, 4)$. The shortest distance between the lines L_1 and L_2 is :

(1) $\frac{23}{\sqrt{38}}$

(2) $\frac{21}{\sqrt{57}}$

(3) $\frac{23}{\sqrt{57}}$

(4) $\frac{21}{\sqrt{38}}$

Q3. Let the shortest distance between the lines $\frac{x-3}{3} = \frac{y-\alpha}{-1} = \frac{z-3}{1}$ and $\frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-\beta}{4}$ be $3\sqrt{30}$. Then the positive value of $5\alpha + \beta$ is

(1) 42

(2) 46

(3) 48

(4) 40

Q4. If the shortest distance between the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x}{1} = \frac{y}{\alpha} = \frac{z-5}{1}$ is $\frac{5}{\sqrt{6}}$, then the sum of all possible values of α is

(1) $\frac{3}{2}$

(2) $-\frac{3}{2}$

(3) 3

(4) -3

Q5. Let the vertices Q and R of the triangle PQR lie on the line $\frac{x+3}{5} = \frac{y-1}{2} = \frac{z+4}{3}$, $QR = 5$ and the coordinates of the point P be $(0, 2, 3)$. If the area of the triangle PQR is $\frac{m}{n}$ then :

(1) $m - 5\sqrt{21}n = 0$

(2) $2m - 5\sqrt{21}n = 0$

(3) $5m - 2\sqrt{21}n = 0$

(4) $5m - 21\sqrt{2}n = 0$

Q6. If the image of the point $P(1, 0, 3)$ in the line joining the points $A(4, 7, 1)$ and $B(3, 5, 3)$ is $Q(\alpha, \beta, \gamma)$, then $\alpha + \beta + \gamma$ is equal to

(1) $\frac{47}{3}$

(2) $\frac{46}{3}$

(3) 18

(4) 13

Q7. Let a line passing through the point $(4, 1, 0)$ intersect the line $L_1; \frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ at the point $A(\alpha, \beta, \gamma)$ and the line $L_2 : x - 6 = y = -z + 4$ at the point $B(a, b, c)$.

Then $\begin{vmatrix} 1 & 0 & 1 \\ \alpha & \beta & \gamma \\ a & b & c \end{vmatrix}$ is equal to

(1) 8

(2) 16

(3) 12

(4) 6

Q8. Line L_1 passes through the point $(1, 2, 3)$ and is parallel to Z -axis. Line L_2 passes through the point $(\lambda, 5, 6)$ and is parallel to y -axis. Let for $\lambda = \lambda_1, \lambda_2, \lambda_2 < \lambda_1$, the shortest distance between the two lines be 3. Then the square of the distance of the point $(\lambda_1, \lambda_2, 7)$ from the line L_1 is

(1) 40

(2) 32

(3) 25

(4) 37

Q9. The distance of the point $(7, 10, 11)$ from the line $\frac{x-4}{1} = \frac{y-4}{0} = \frac{z-2}{3}$ along the line $\frac{x-9}{2} = \frac{y-13}{3} = \frac{z-17}{6}$ is

- (1) 18
- (2) 14
- (3) 12
- (4) 16

Q10. Let A and B be two distinct points on the line $L : \frac{x-6}{3} = \frac{y-7}{2} = \frac{z-7}{-2}$. Both A and B are at a distance $2\sqrt{17}$ from the foot of perpendicular drawn from the point $(1, 2, 3)$ on the line L. If O is the origin, then $\vec{OA} \cdot \vec{OB}$ is equal to:

- (1) 49
- (2) 47
- (3) 21
- (4) 62

Q11. Let the values of p, for which the shortest distance between the lines $\frac{x+1}{3} = \frac{y}{4} = \frac{z}{5}$ and $\vec{r} = (p\hat{i} + 2\hat{j} + \hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 4\hat{k})$ is $\frac{1}{\sqrt{6}}$, be a, b, ($a < b$). Then the length of the latus rectum of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is :-

- (1) 9
- (2) $\frac{3}{2}$
- (3) $\frac{2}{3}$
- (4) 18

Q12. Let A be the point of intersection of the lines $L_1 : \frac{x-7}{1} = \frac{y-5}{0} = \frac{z-3}{-1}$ and $L_2 : \frac{x-1}{3} = \frac{y+3}{4} = \frac{z+7}{5}$. Let B and C be the point on the lines L_1 and L_2 respectively such that $AB = AC = \sqrt{15}$. Then the square of the area of the triangle ABC is :

- (1) 54
- (2) 63
- (3) 57
- (4) 60

Q13. Let the line L pass through $(1, 1, 1)$ and intersect the lines $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{4}$ and $\frac{x-3}{1} = \frac{y-4}{2} = \frac{z}{1}$. Then, which of the following points lies on the line L ?

- (1) $(4, 22, 7)$
- (2) $(5, 4, 3)$
- (3) $(10, -29, -50)$
- (4) $(7, 15, 13)$

Q14. Consider the lines $L_1 : x - 1 = y - 2 = z$ and $L_2 : x - 2 = y = z - 1$. Let the feet of the perpendiculars from the point $P(5, 1, -3)$ on the lines L_1 and L_2 be Q and R respectively. If the area of the triangle PQR is A , then $4A^2$ is equal to :

- (1) 139
- (2) 147
- (3) 151
- (4) 143

Q15. Let the values of λ for which the shortest distance between the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-\lambda}{3} = \frac{y-4}{4} = \frac{z-5}{5}$ is $\frac{1}{\sqrt{6}}$ be λ_1 and λ_2 . Then the radius of the circle passing through the points $(0, 0)$, (λ_1, λ_2) and (λ_2, λ_1) is

- (1) $\frac{5\sqrt{2}}{3}$
- (2) 4
- (3) $\frac{\sqrt{2}}{3}$
- (4) 3

Q16. Let the area of the triangle formed by the lines $x + 2 = y - 1 = z$, $\frac{x-3}{5} = \frac{y}{-1} = \frac{z-1}{1}$ and $\frac{x}{-3} = \frac{y-3}{3} = \frac{z-2}{1}$ be A . Then A^2 is equal to _____

Q17. If the equation of the line passing through the point $(0, -\frac{1}{2}, 0)$ and perpendicular to the lines

$$\vec{r} = \lambda(\hat{i} + a\hat{j} + b\hat{k}) \text{ and}$$

$$\vec{r} = (\hat{i} - \hat{j} - 6\hat{k}) + \mu(-b\hat{i} + a\hat{j} + 5\hat{k})$$

is $\frac{x-1}{-2} = \frac{y+4}{d} = \frac{z-c}{-4}$, then $a + b + c + d$ is equal to :

- (1) 10
- (2) 14
- (3) 13

